
Fruit tea sales analysis and forecasting

BMAN71791 Applied Statistics and Business Forecasting

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Data Exploration

2

Strategic Insights:
Navigating Key Questions

3

Limitations

4

Conclusion and
Recommendations

1

Data Exploration

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Strategic Insights:
Navigating Key Questions

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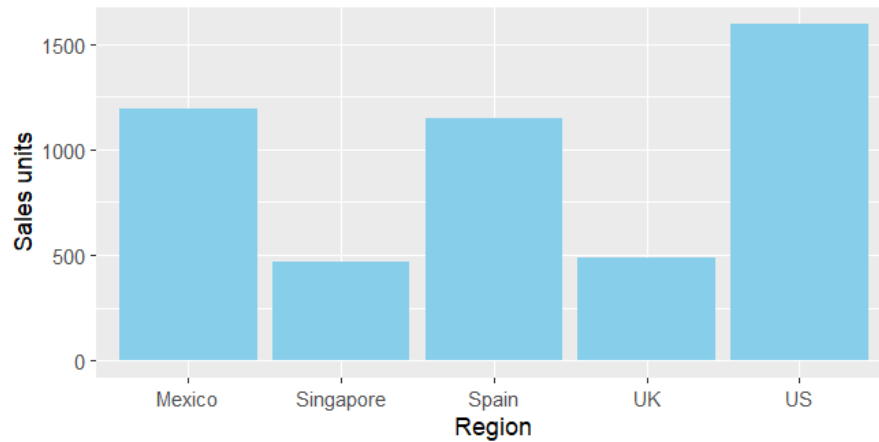
Limitations

4

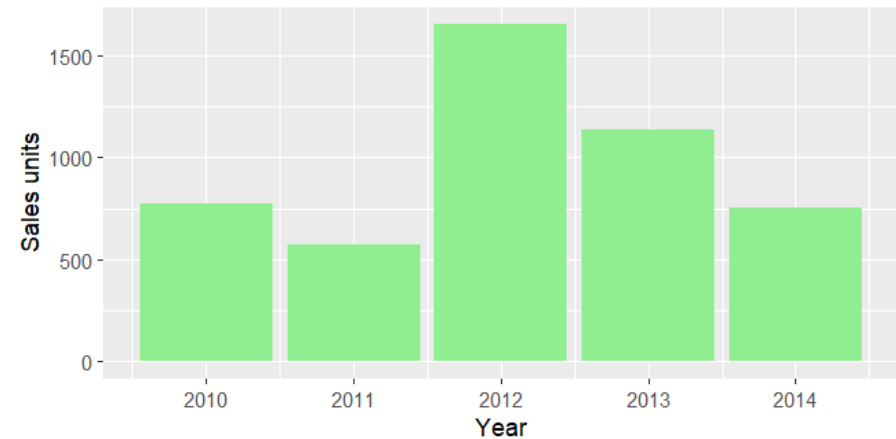
Conclusion and
Recommendations

Data Exploration

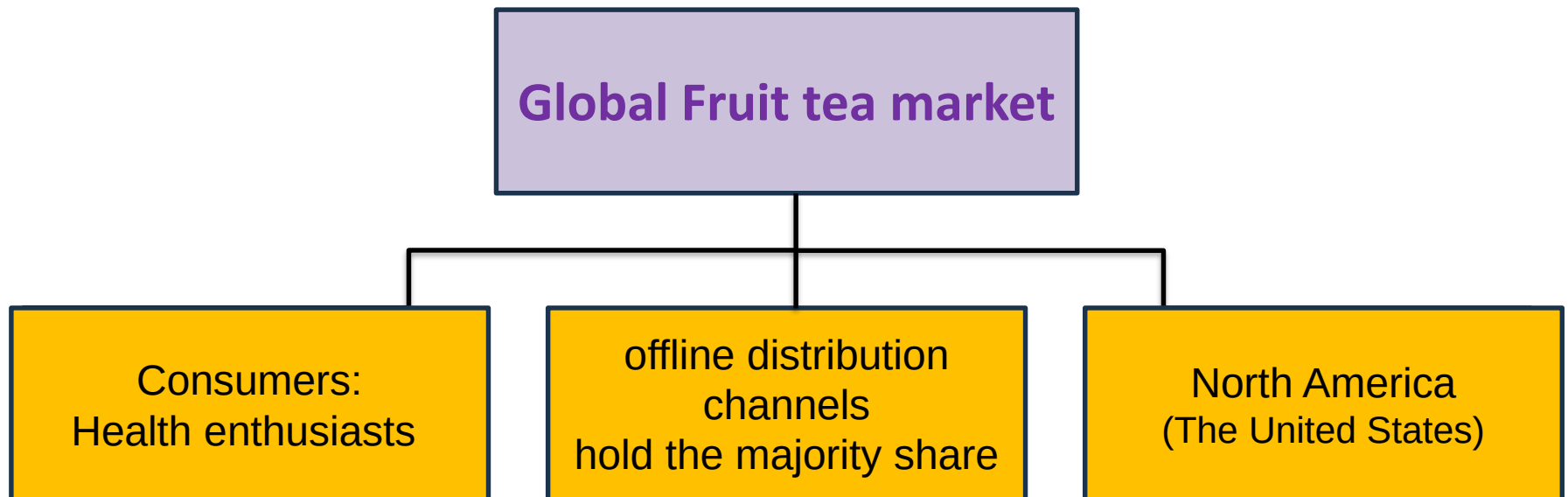
Sales in different Region



Sales in different Year



Business Background



Correlation matrix

	Sales	Price	Ad1	Ad2	Promotions	Wage (%)	Time
Sales	1.000	0.037	0.102	0.694	0.005	0.047	0.046
Price	0.037	1.000	-0.034	0.006	-0.021	0.005	0.005
Ad1	0.102	-0.034	1.000	0.107	-0.096	0.217	0.217
Ad2	0.694	0.006	0.107	1.000	0.036	-0.090	-0.090
Promotions	0.005	-0.021	-0.096	0.036	1.000	0.141	0.140
Wage (%)	0.047	0.005	0.217	-0.090	0.141	1.000	1.000
Time	0.046	0.005	0.217	-0.090	0.140	1.000	1.000

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Data Preparation

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**Strategic Insights:
Navigating Key Questions**

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Limitations

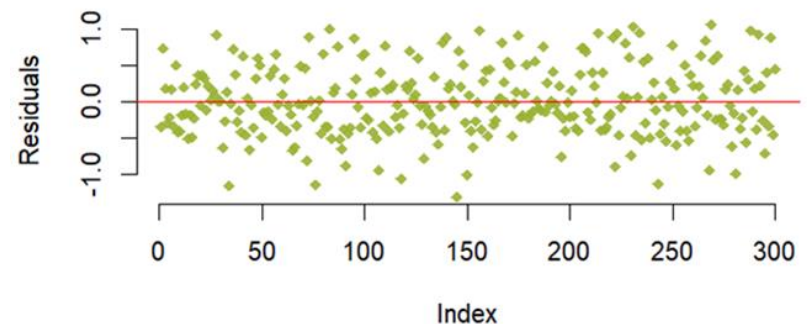
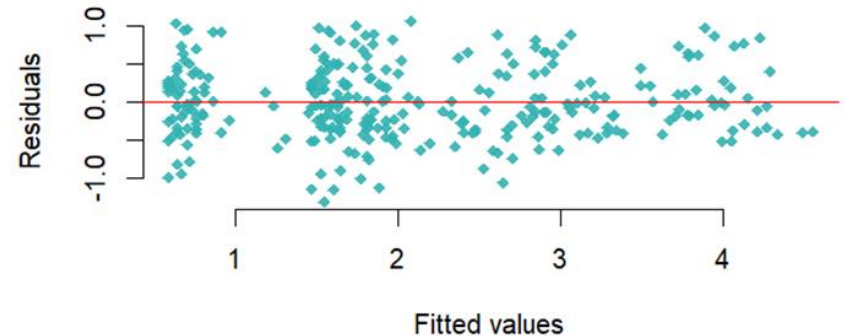
4

Conclusion and
Recommendations

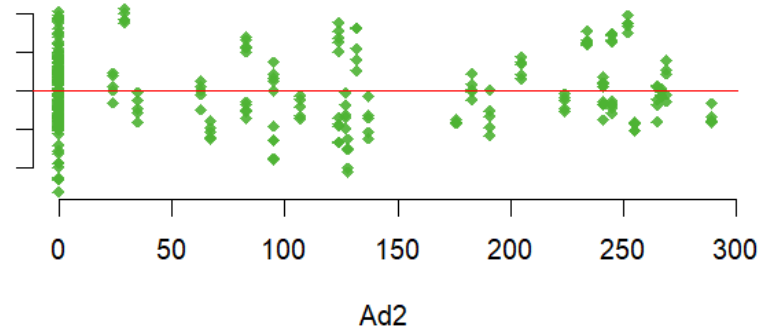
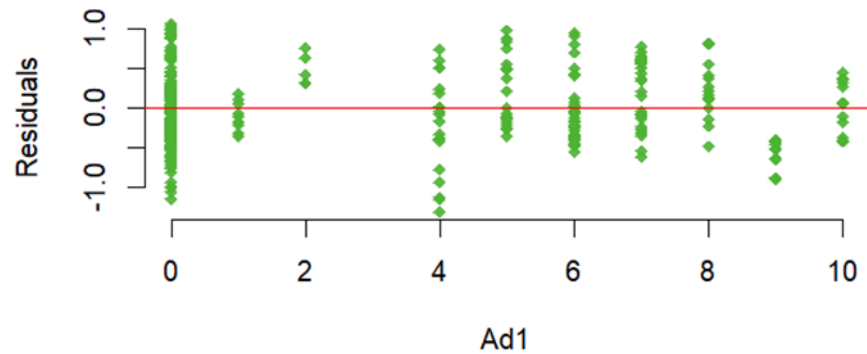
Question 1

Linear regression assumptions on residuals:

- ☐ Independence
- ☐ Expected value of Residuals = 0
- ☐ Homoscedasticity (have the same variance)
- ☐ Residuals follow a normal distribution



Question 1



Ad 1

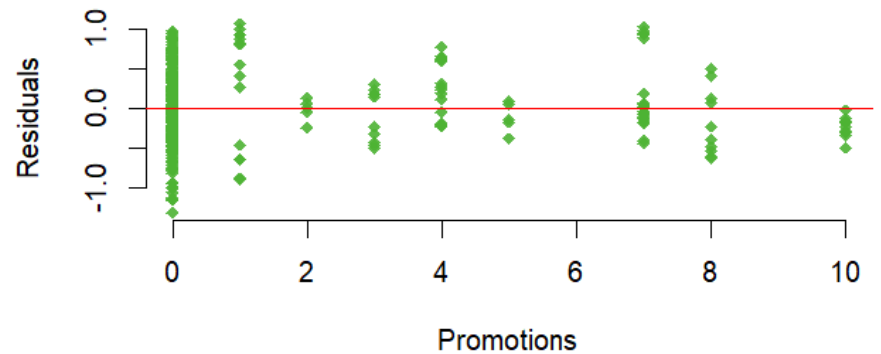
TV Advertisements

Ad2

Online banner ads

Promotions

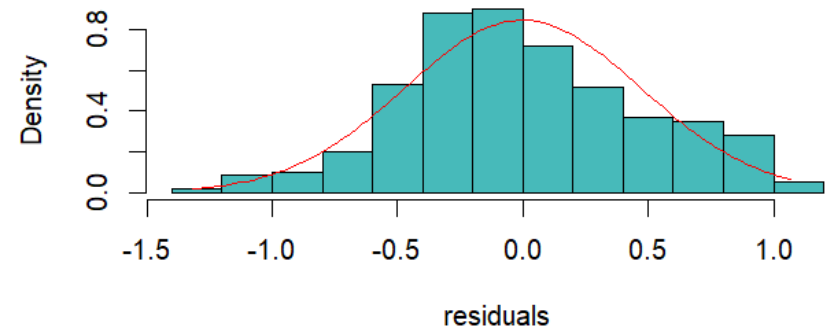
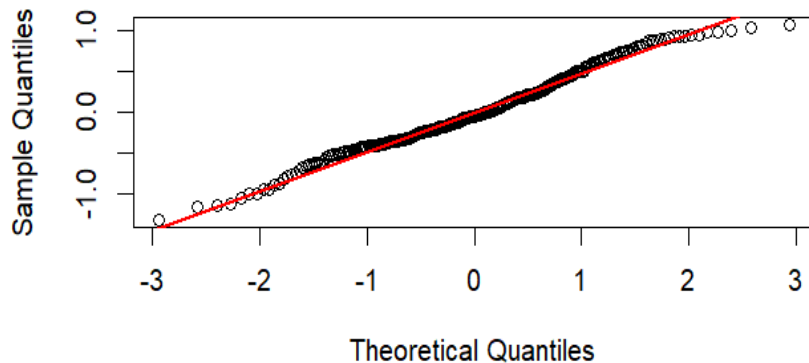
Promotion at distribution stores



Question 1

Linear Relation between Sales and Ads

Normal Q-Q Plot

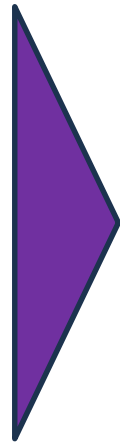


Advertisement	Times rise in units sold
1 unit increase in Ad1 (TV advertisements)	1.0196
1 unit increase in Ad2 (No. of banners)	1.009252
1 unit increase in Ad3 (Promotions)	1.006325

Question 2

TV cost
= £2,000,000

Banners cost
= £500,000



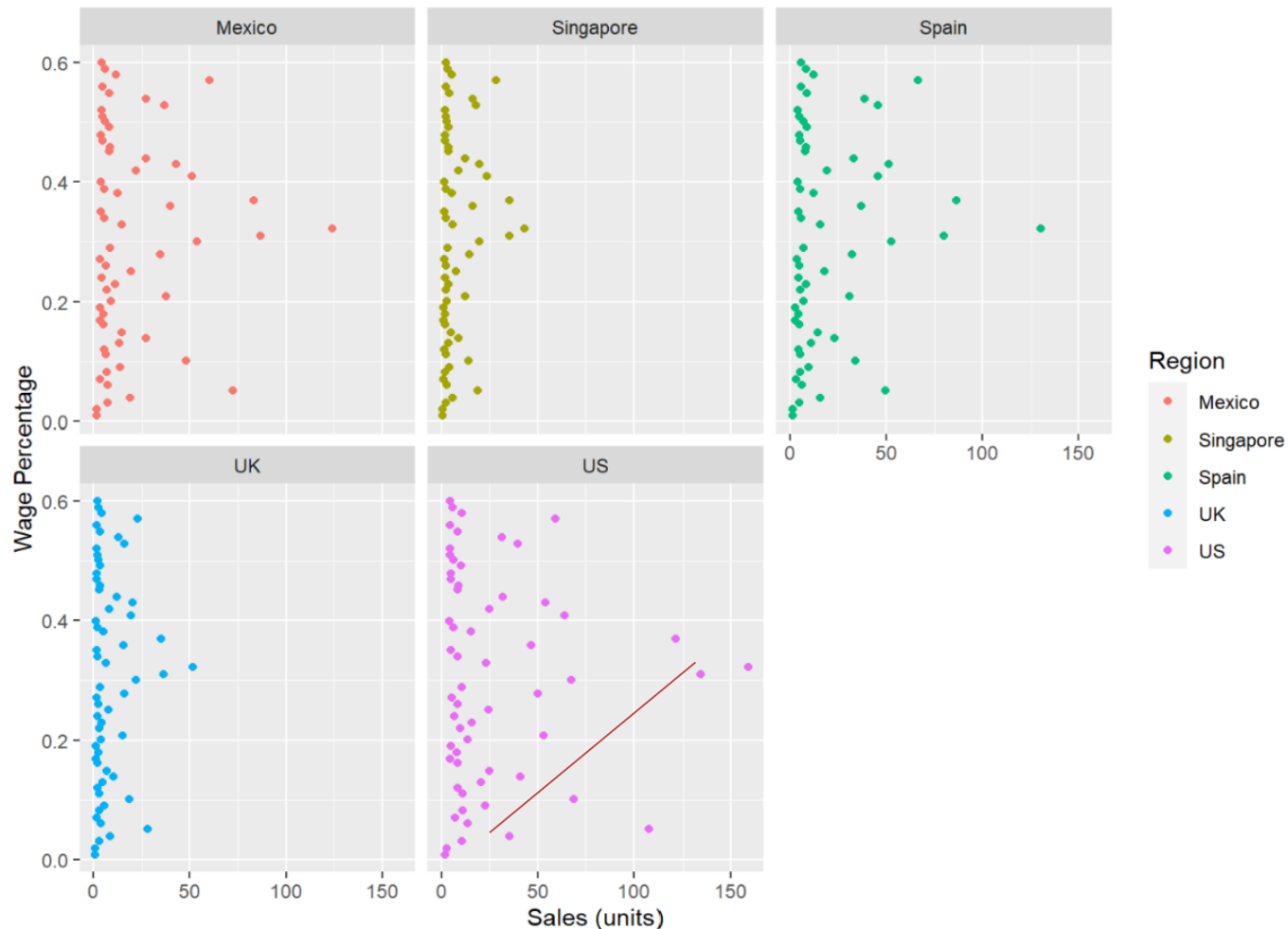
Coefficient*0.3
/cost



Banners is
cost-effective

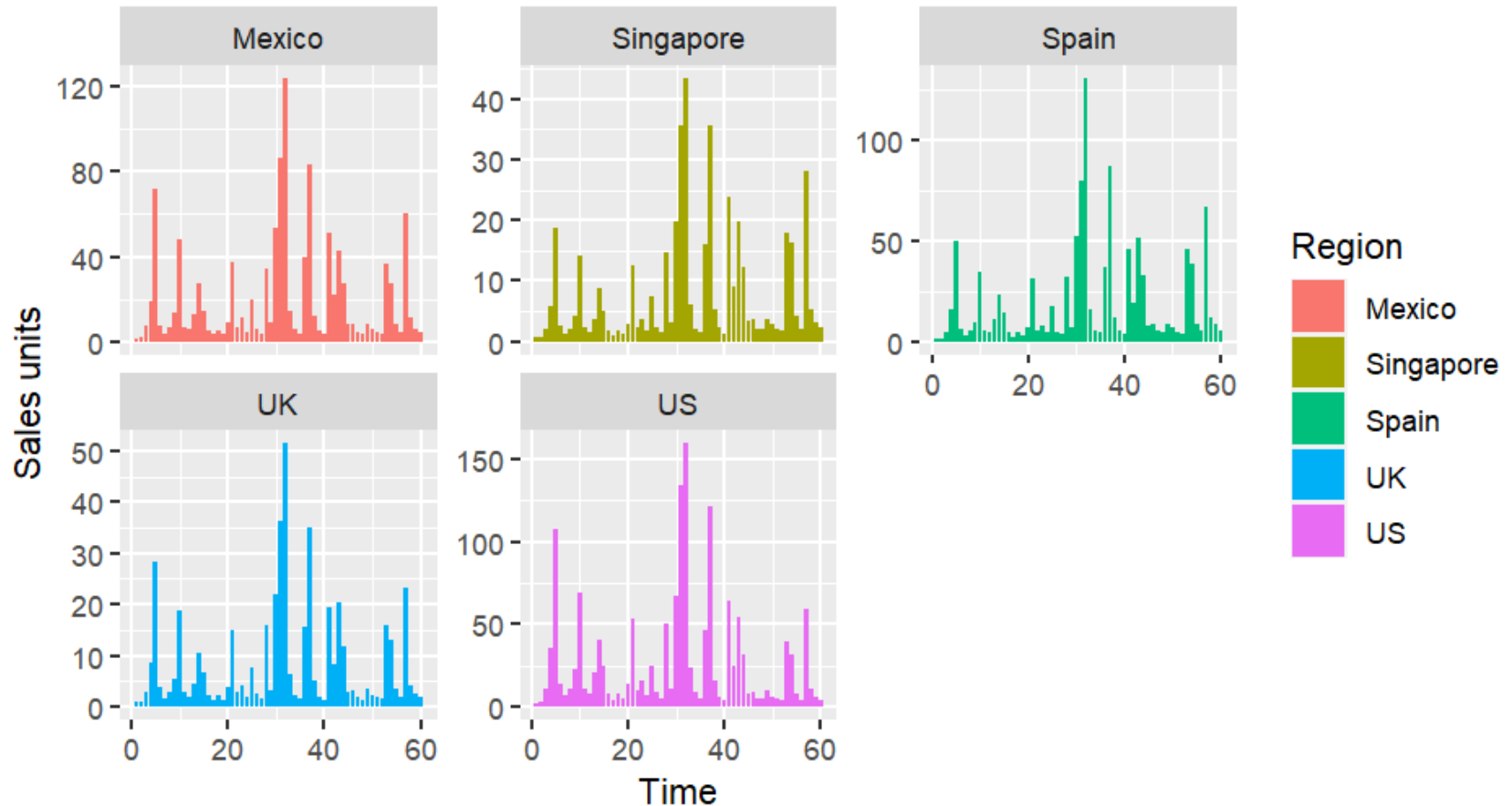
Question 3 : Wage percentage

Relationship between Sales, Wage Percentage, and Region



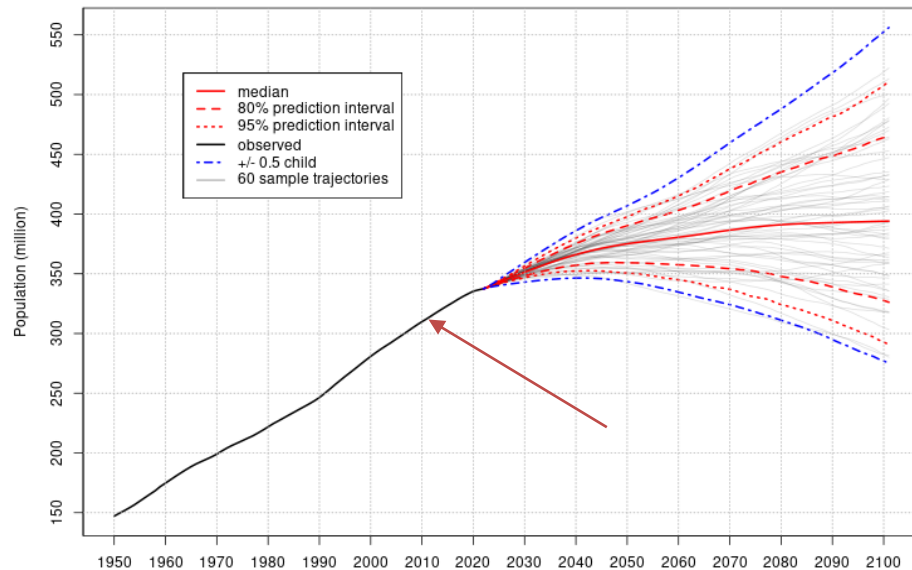
Question 3 : Population

Sales in Different Regions in Different Time



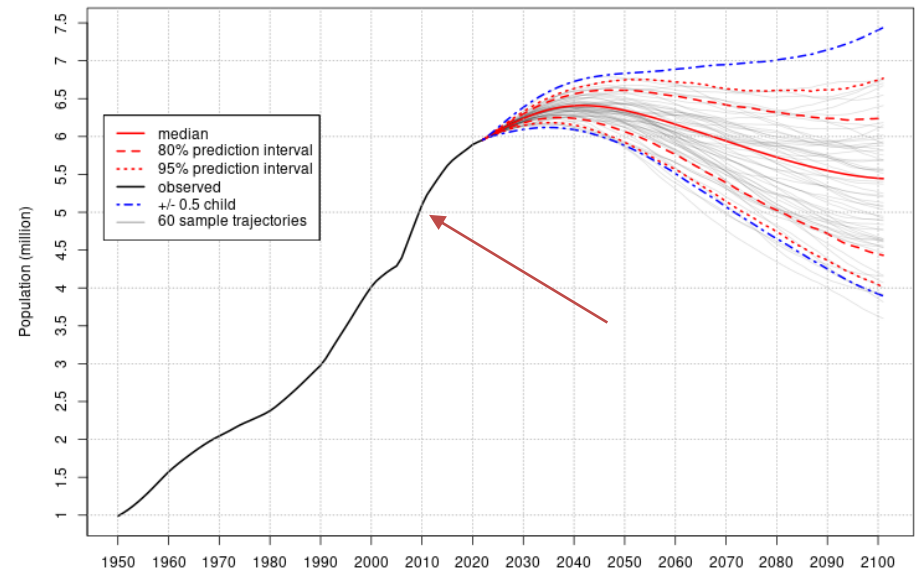
Question 3 : Population

United States of America: Total Population



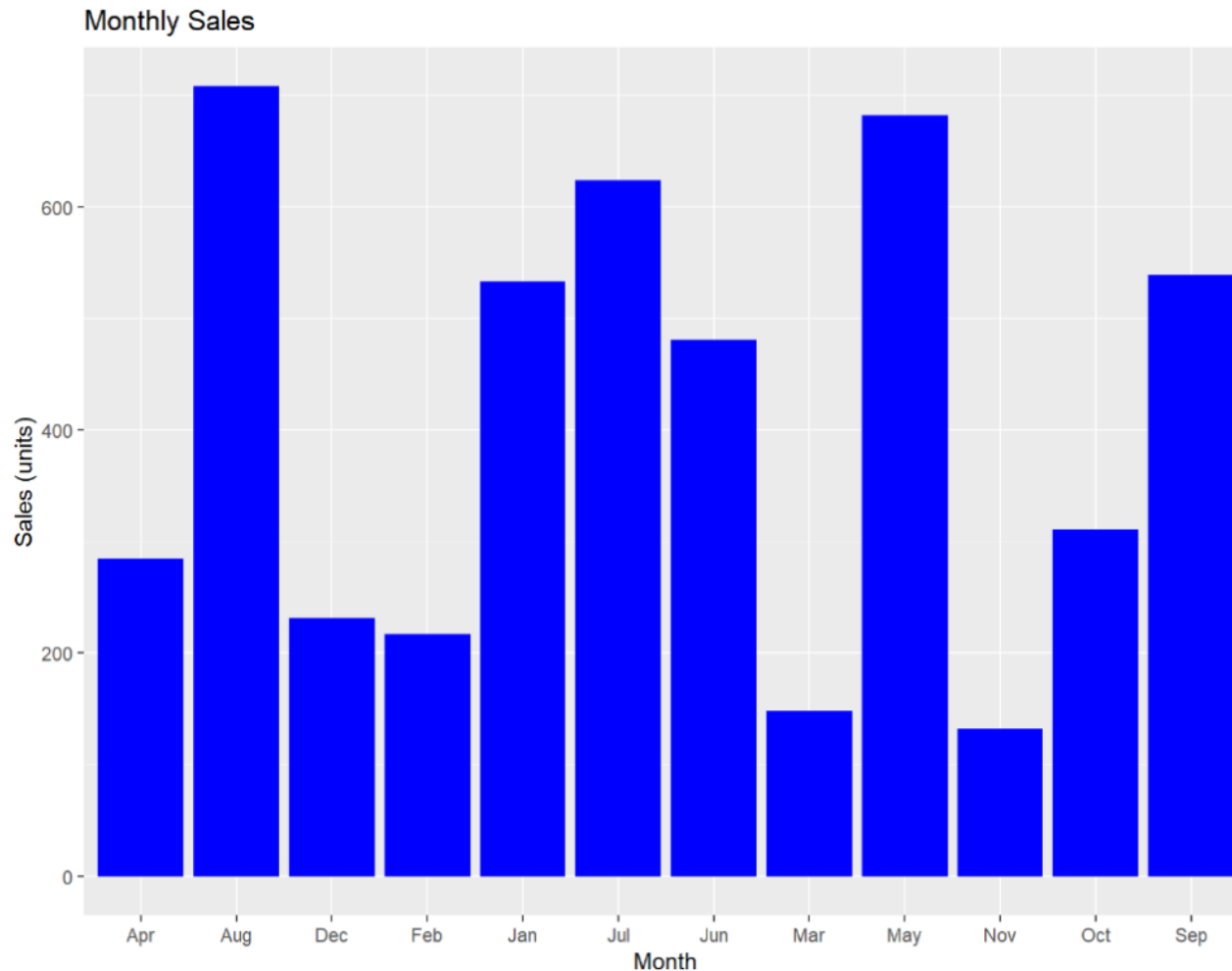
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United Nations, DESA, Population Division. *World Population Prospects 2022*. <http://population.un.org/wpp/>

Singapore: Total Population



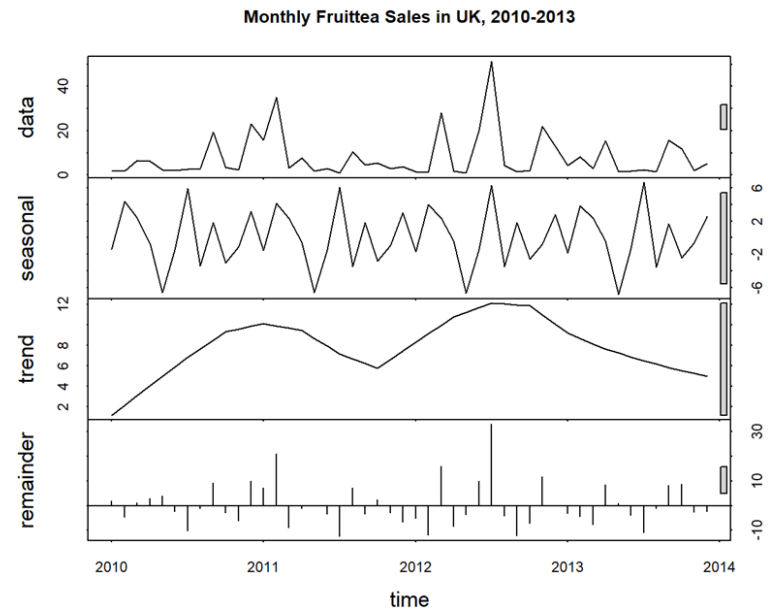
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United Nations, DESA, Population Division. *World Population Prospects 2022*. <http://population.un.org/wpp/>

Question 3 : Seasonal Trends

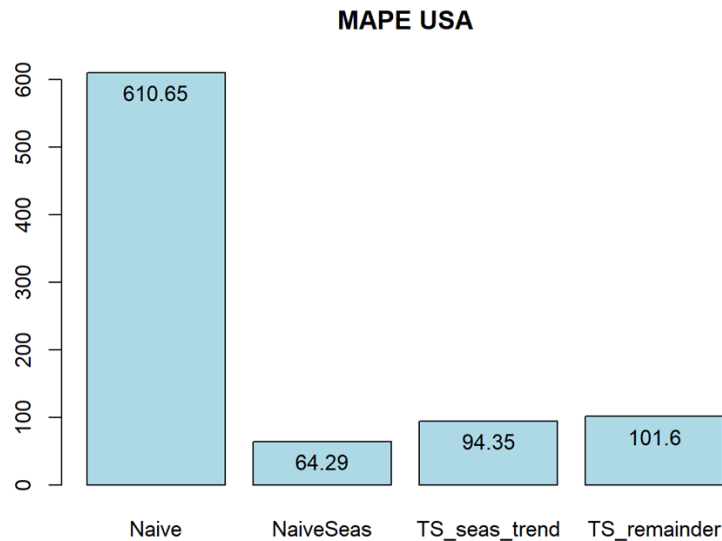


Question 4

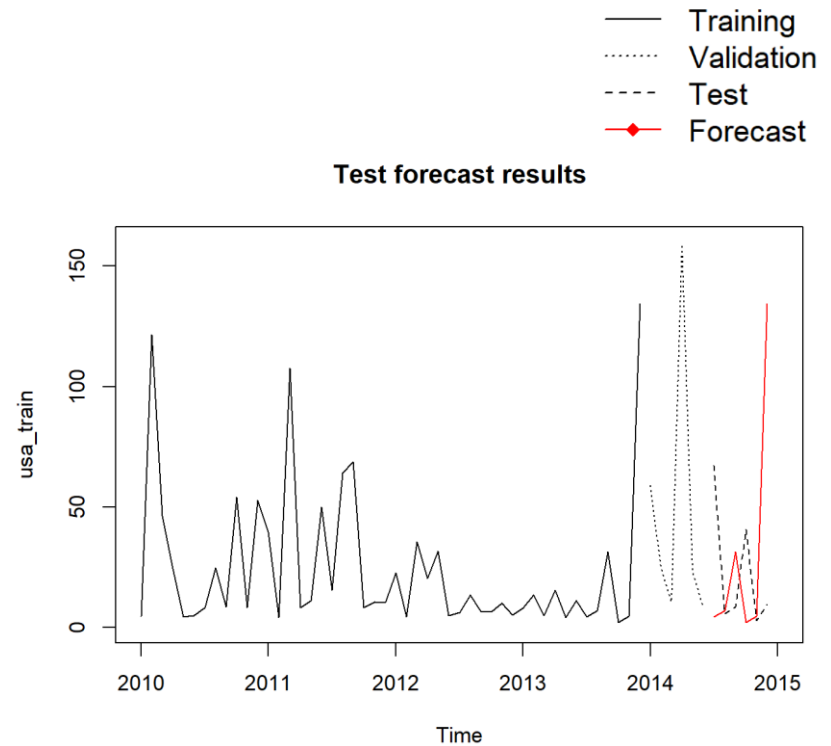
- Decomposition of the data shows clear seasonal pattern but non-linear trend



USA validation and test results

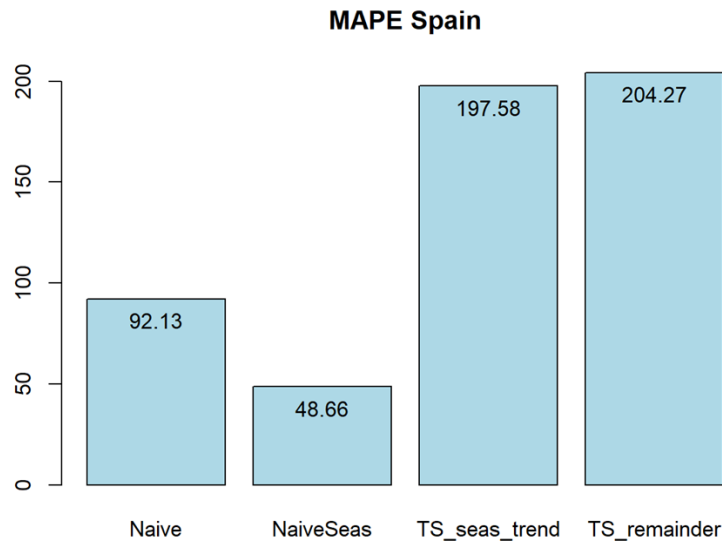


MAPE values for different forecasting models on validation set

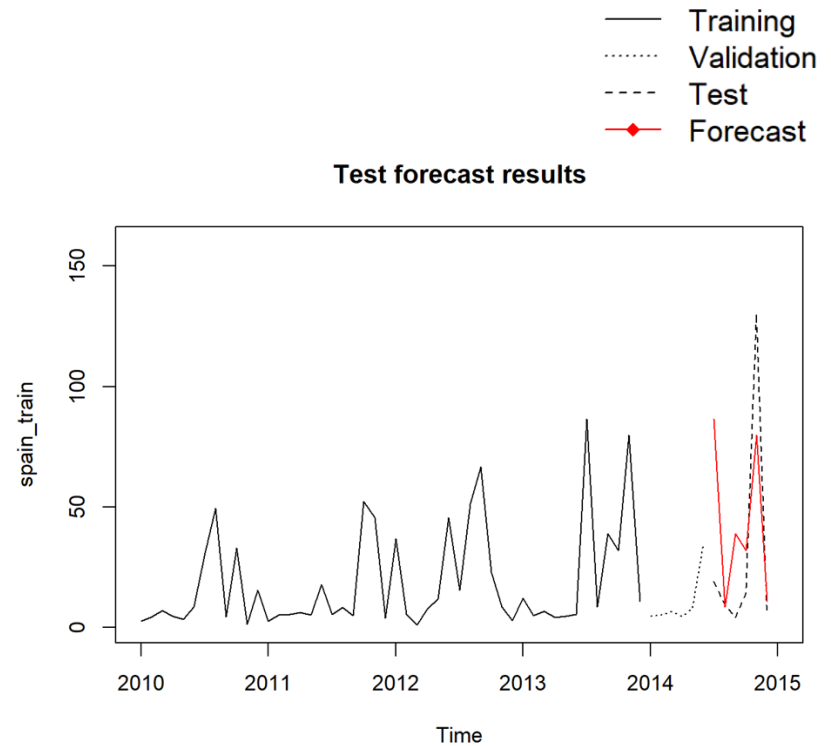


Forecast for test set using Naive Seasonality model

Spain validation and test results

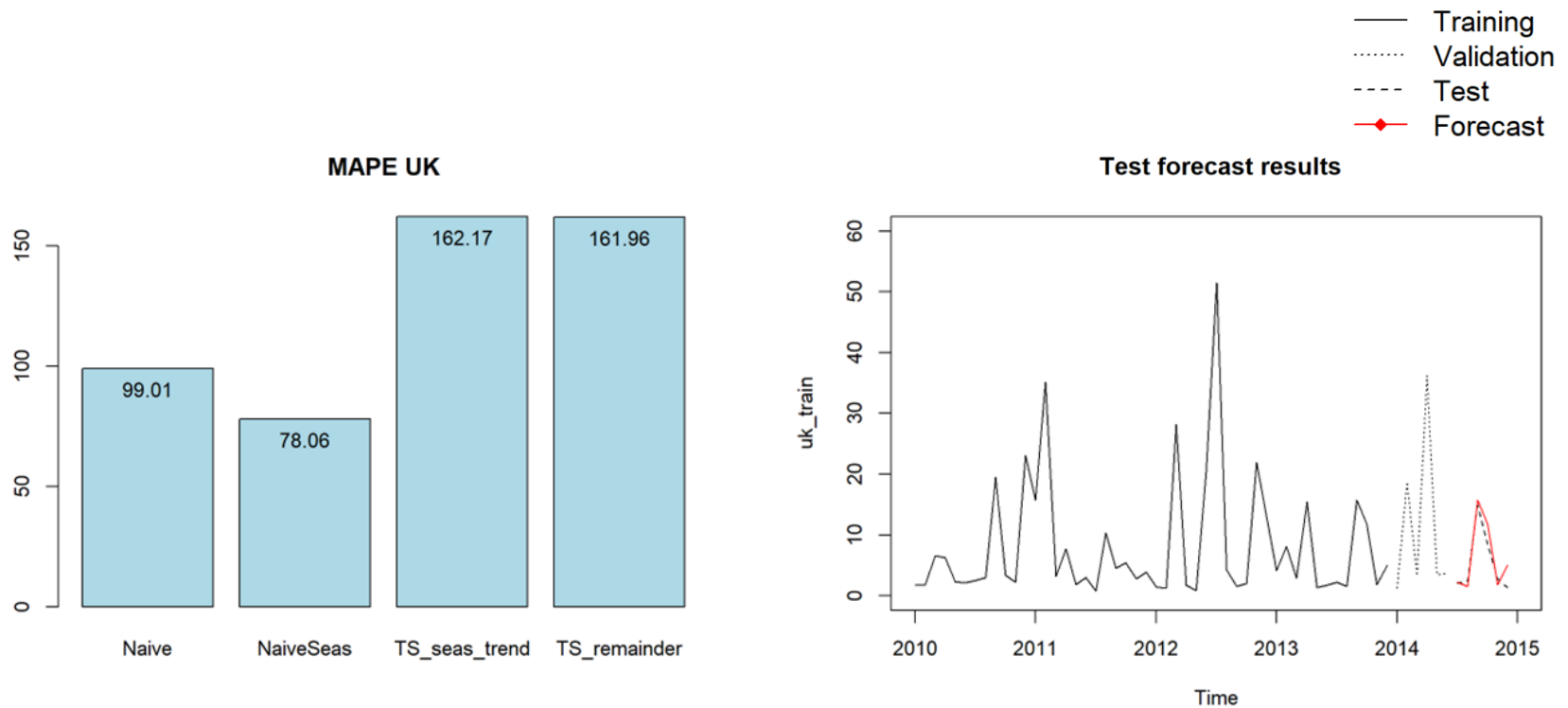


MAPE values for different forecasting models on validation set



Forecast for test set using Naive Seasonality model

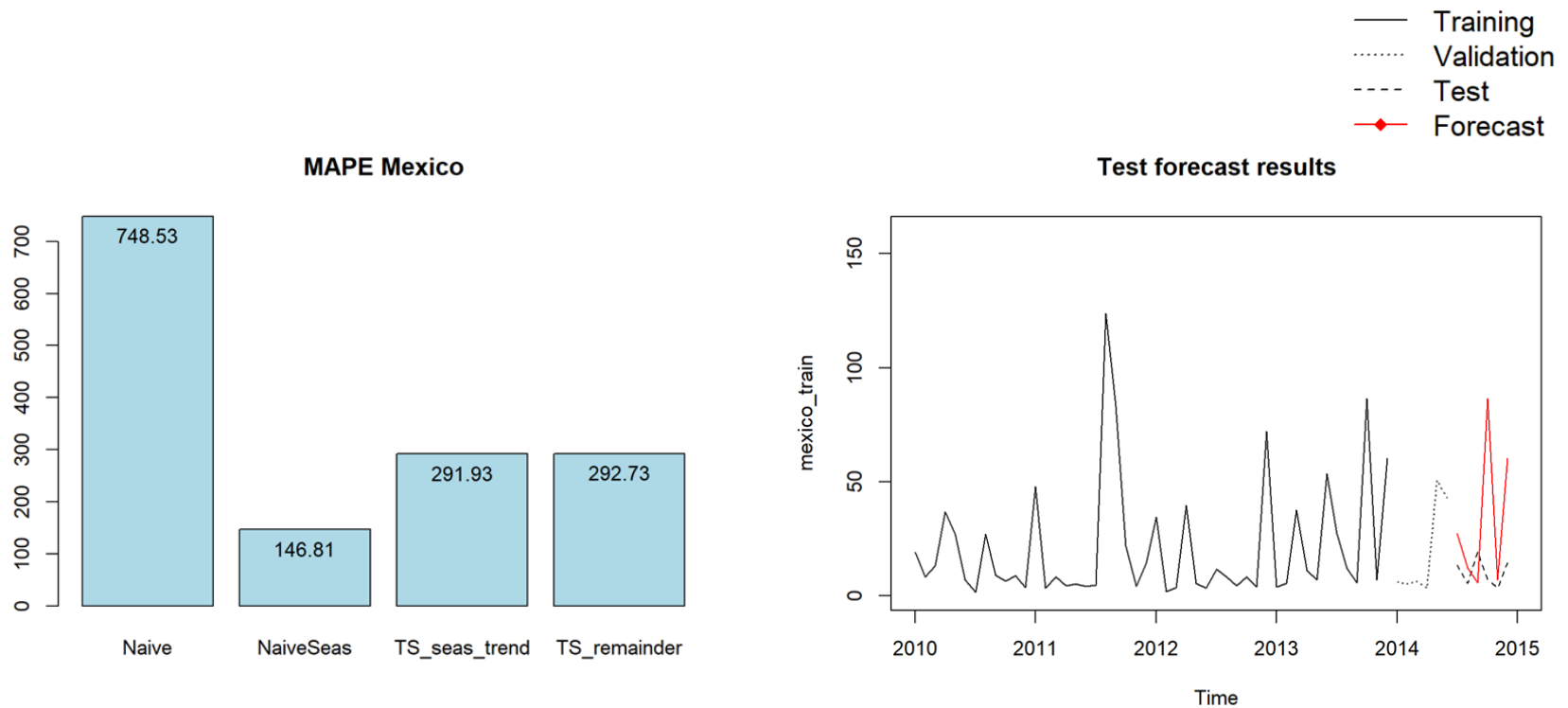
UK validation and test results



MAPE values for different forecasting models on validation set

Forecast for test set using Naïve Seasonality model

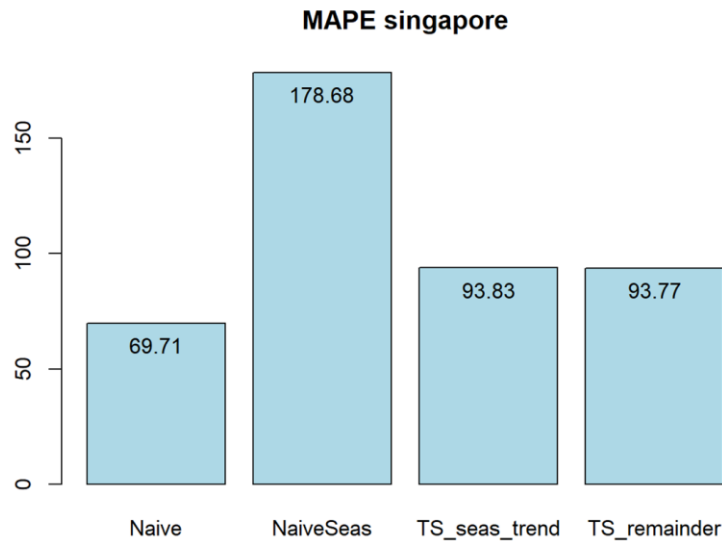
Mexico validation and test results



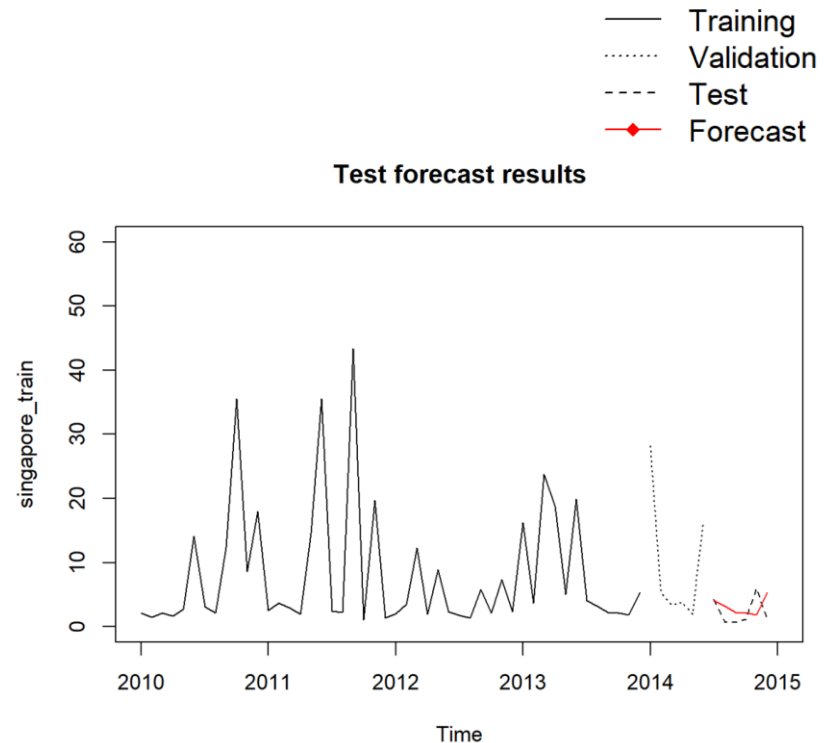
MAPE values for different forecasting models on validation set

Forecast for test set using Naïve Seasonality model

Singapore validation and test results



MAPE values for different forecasting models on validation set



Forecast for test set using Naive Seasonality model

Forecasted values

Method used :
Naïve Seasonal
Forecasting

Comment : Though the stl function shows a simple Additive decomposition for all 5 regions, forecasting using decomposition shows poor results, possibly due to non-linear trend

Region	Forecasted unit Sales for Jan-2015
USA	8.08
Spain	12.18
UK	4.22
Mexico	3.9
Singapore	16.28

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Data Preparation

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Strategic Insights:
Navigating Key Questions

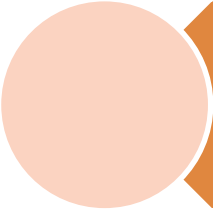
3

Limitations

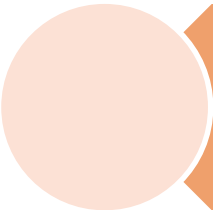
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Conclusion and
Recommendations

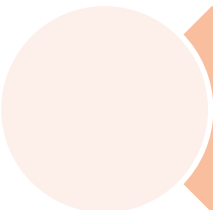
Limitations



Limited Dataset



Numerous Zero Observations



Customer Distribution

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Recommendations

Conclusion & Recommendations

Increase sales

Add more
advertisement in
winter

Reduce cost

Decrease the
use of store
promotion(Ad3)

Thank you